

Exploratory Data Analysis

6.1 Market Price Analysis

Analysed:

- Average price by city
- Room type differences
- Neighbourhood attraction
- Price distribution

Key observations:

- Sydney, Cape Town, and New York are the top three cities that have the highest prices, while Istanbul and Bangkok offered more affordable accommodation. The differences in currency are one of the most impactful reasons for this phenomenon.
- Entire properties and hotel rooms generally commanded premium prices compared with shared rooms and private rooms.

6.2 Seasonality Analysis

Demand Analysis

The yearly demand trend analysis was conducted by integrating Airbnb listing information with review data using the listing ID as the connection key. This allowed review activity to be analysed at different time periods and across different geographical markets. Review dates were transformed into yearly and monthly components, with demand evaluated based on the number of reviews received over time. Since reviews represent completed stays and traveller engagement, the monthly review count was used as a proxy indicator for Airbnb demand.

The analysis focused on identifying major changes in market activity, including peak tourism periods, low-demand periods, the impact of the COVID-19 pandemic, and the subsequent recovery period. Before COVID-19, most cities demonstrated relatively stable or increasing review activity, reflecting continuous growth in short-term rental demand and tourism activity. However, a significant decline in review volumes was observed during the COVID-19 period due to international travel restrictions, reduced mobility, and disruptions to the tourism industry. This decline highlights the vulnerability of Airbnb markets to external events that affect traveller behaviour.

Following the relaxation of travel restrictions, review activity gradually increased across cities, indicating a recovery in tourism demand. However, the speed and strength of recovery differed

between locations, with some cities returning to previous demand levels faster due to stronger tourism infrastructure, international popularity, and domestic travel demand. By comparing yearly demand patterns between cities, the analysis identified city-specific recovery trends and provided insights into market resilience after major disruptions.

Seasonality

Seasonal demand analysis was performed by grouping review data according to **month, year, and city** to identify recurring travel patterns and fluctuations in Airbnb demand throughout the year.

Monthly review counts were analysed to determine when traveller activity increased or decreased, allowing the identification of peak tourism seasons and low-demand periods for each market.

The analysis revealed that Airbnb demand is influenced by seasonal travel behaviour, with different cities experiencing different demand cycles. During peak tourism seasons, review activity increased as more travellers visited popular destinations, leading to higher occupancy potential and stronger revenue opportunities for property owners. In contrast, low seasons were associated with reduced traveller activity, which may result in lower occupancy rates and increased competition among hosts.

By comparing monthly demand patterns across cities, the analysis identified city-specific seasonal characteristics. Some destinations experienced strong seasonal peaks during traditional holiday periods or warmer months, while others maintained more consistent demand throughout the year due to business travel, cultural attractions, or diverse tourism activities. Understanding these seasonal variations is important for investors and Airbnb hosts because it supports better decisions regarding pricing strategies, property availability, and market entry timing.

6.3 Review rating analysis

A correlation analysis was conducted to investigate the relationship between the overall Airbnb review rating (**review_scores_rating**) and individual review categories, including accuracy, cleanliness, check-in experience, communication, and location. The purpose of this analysis was to identify which aspects of the guest experience have the strongest influence on the overall review rating. The correlation matrix was calculated using the Pearson correlation coefficient, where values closer to **1** indicate a strong positive relationship, while values closer to **0** indicate a weak relationship.

The results demonstrate that all review categories have a positive relationship with the overall review rating, indicating that improvements across different aspects of the Airbnb experience contribute to higher guest satisfaction.

Among all factors, **review accuracy** has the strongest correlation with the overall rating, with a coefficient of **0.786**. This suggests that accurately representing the property information, including

descriptions, amenities, and images, is one of the most important factors affecting guest satisfaction. When the actual experience matches guests' expectations, they are more likely to provide higher overall ratings.

Cleanliness also showed a strong positive correlation with overall rating (**0.737**). This highlights that maintaining a clean and comfortable environment is a critical factor in determining guest satisfaction. Poor cleanliness can negatively affect the overall experience, even when other aspects of the accommodation are satisfactory.

The correlation between **communication** and overall rating was **0.714**, demonstrating the importance of effective host-guest interactions. Timely responses, clear instructions, and helpful support can improve the guest experience and contribute to higher ratings.

Check-in experience had a moderate-to-strong correlation with overall rating (**0.667**). This indicates that a smooth arrival process, including convenient access and clear check-in procedures, positively influences guests' impressions of the property.

Among all factors, **location** had the lowest correlation with overall rating (**0.551**). Although location remains an important consideration when selecting accommodation, the results suggest that service-related factors, such as accuracy, cleanliness, and communication, have a stronger influence on guests' final ratings.

Furthermore, the review attributes also showed positive relationships with each other. For example, check-in and communication had a correlation of **0.744**, suggesting that effective communication can contribute to a smoother check-in experience.

Overall, the analysis indicates that **accuracy, cleanliness, and communication are the most influential factors associated with Airbnb review ratings**. These findings suggest that hosts seeking to improve their overall ratings should focus on providing accurate property information, maintaining high cleanliness standards, and ensuring effective communication with guests throughout their stay.

6.4 Value-for-Money Analysis

The objective of this analysis is to answer the question: **“Which city provides the best travel experience considering both review quality and price?”** A city with a high value score represents a destination where visitors can obtain higher-quality accommodation experiences at a relatively affordable price.

The analysis involved several steps. First, Airbnb listings were grouped by city to calculate the average **review_scores_rating** and average price for each location. Next, the review ratings and prices were standardised to ensure fair comparison between cities. Finally, the Value-for-Money index was calculated and used to rank cities based on accommodation affordability and quality.

The results provide practical recommendations for travellers by identifying cities that offer the best combination of price and guest experience. Cities with high review scores but extremely high prices may not necessarily provide the best value, while cities with moderate prices and consistently positive reviews may achieve higher value-for-money rankings.

This analysis demonstrates how Airbnb data can be transformed into actionable travel insights. Instead of simply selecting the cheapest destination or the highest-rated accommodation, travellers can use the Value-for-Money index to make more informed decisions by balancing **cost efficiency and accommodation quality**. For Airbnb hosts and businesses, the analysis also highlights opportunities to improve competitiveness by understanding how pricing strategies relate to guest satisfaction.